

prompting iterations, allowing our method to generate an unlimited length of stories with increased levels of storyline complexity.

3.3 Detail Elaboration

With the action-reason pairs obtained in subsection 3.2, we convert each pair into a short summary for a story chapter in which the protagonist tries to take the given action but that action does not work due to the given reason. Conditioned on the short summary, we prompt the LLM to generate a sequence of events that can serve as an outline for each story chapter. During prompting we need to make sure the protagonist is not aware that their plan will fail (readers assume the protagonist is rational and will not take actions that they believe are unlikely to be effective in achieving their goal). With a sequence of events, the full story is then generated by asking the LLM for an elaboration on the event details.

Aiming to further improve the level of suspense readers could possibly perceive, we incorporate two optional designs into our method for suspenseful story generation. (1) we ask the LLM to set up some small clues for the upcoming failure of the protagonist’s action so as to better arouse readers’ anticipation of the protagonist’s failure. (2) We draw inspiration from the narrative theory of *disparity of knowledge* (Branigan, 2013) and design two information revelation modes. In the first, we reveal the reason the protagonist’s actions will fail *beforehand*, setting up an anticipation of failure in the reader that the protagonist doesn’t know about and is thus unlikely to be able to avert. For example, if our protagonist is a spy, they might be unaware that the antagonists have set a trap. In the second, we reveal the information *after the fact*, explaining to the reader how the protagonist’s actions were doomed to fail from the beginning.

A complete run of our method starts with §3.1 Background Setup and may contain two or more prompting iterations. Each of the prompting iterations comprises §3.2 Outline Planning and §3.3 Detail Elaboration, which are repeated in our method to create stories in which the protagonist repeatedly fails to prevent the dire situation. In the final iteration, we do not generate the failure response (i.e., step #2 of subsection 3.2), and we allow the protagonist to succeed in preventing the dire situation and achieving the goal. The prompts we use in the detail elaboration stage and an example of LLM’s corresponding outputs are presented in Ap-

Protagonist’s Name and Occupation: \$name_occupation Protagonist’s Goal: \$goal Protagonist’s Dire Situation (if the goal is not achieved): \$dire_situation Tell me about a concrete action the protagonist is most likely to take, in order to achieve the goal. <i>Max Sterling, employing their extensive network of contacts and utilizing their advanced tech skills, infiltrates a top-secret meeting of high-ranking Phoenix members, planting surveillance devices to gather valuable intel on the organization’s plans and the whereabouts of its leader. (We denote the model output as \$action1.)</i>
The protagonist tries to perform the following action: \$action1. But the protagonist still fails to achieve the goal for a reason. Tell me about what this reason could be. <i>Despite Max Sterling’s meticulous planning and execution, they are betrayed by a double agent within their own agency, who tips off the Phoenix leaders about the surveillance devices, resulting in the immediate evacuation of the meeting and thwarting Max’s attempt to gather crucial information and track down the organization’s leader. (We denote the model output as \$reason1.)</i>
The protagonist has taken all the following actions: [\$action1]. But it turns out that those actions are not effective in helping the protagonist to achieve the goal, so the goal is still not achieved yet. Tell me about a concrete new action the protagonist is most likely to take next, in order to achieve the goal. <i>Max Sterling decides to track down and approach a former member of “The Phoenix”, who defected years ago but still holds valuable information about the organization and its leader. Using their negotiating skills and persuasive tactics, Max aims to convince the defector to cooperate and provide crucial insider details that will help bring down the terrorist organization once and for all. (We denote the model output as \$action2.)</i>
The protagonist tries to perform the following action: \$action2. But the protagonist still fails to achieve the goal for a reason. Tell me about what this reason could be. <i>Despite Max’s best efforts, the defector is not willing to collaborate, fearing for their own safety and the safety of their loved ones. The defector is haunted by past experiences with “The Phoenix” and their leader, making them reluctant to trust anyone, even someone as skilled and determined as Max. (We denote the model output as \$reason2.)</i>
The protagonist has taken all the following actions: [\$action1, \$action2]. But it turns out ... achieve the goal. (similar to the prompt used for \$action2) <i>Realizing that their previous actions have not yielded the desired results, Max Sterling decides to gain access to The Phoenix’s encrypted communication channels, using their expertise in cryptography and hacking to unravel the organization’s hidden communications and identify potential weak links within the group who could provide vital information about the leader’s location. (We denote the model output as \$action3.)</i>

Table 2: The prompts we use in subsection 3.2 Outline Planning (highlighted in **bold**), and an example of LLM’s corresponding outputs (highlighted in *italic*). All the variables shaped as \$variable are obtained in subsection 3.1, as reflected in Table 1.

pendix B. The corresponding full story is presented in Appendix C.

4 Evaluation

Human participant evaluation has been generally held as the gold-standard evaluation technique for creative language generation tasks such as story generation (Celikyilmaz et al., 2020; Caglayan et al., 2020; van der Lee et al., 2021). To extensively evaluate our method, we design three human participant studies. In the first, participants are asked to judge the goodness of the stories we generate and compare stories generated by our technique against those generated by the baseline method. The comparisons are based on multiple evaluation criteria. In the second, we perform ablations on our system and compare our full system against the ablations. Ablation studies help us understand whether different parts of our system are significant to the solution. In the third, participants are requested to examine the story outline to help determine if the internal workflow is running properly as we expected.

All human participant studies are performed using the Prolific crowdsourcing platform². These studies have been approved by our institution’s Institutional Review Board (IRB). We qualify participants by first asking them a screening question at the beginning of the survey, and then verifying answers manually to disregard answers provided by those who fail the screening. We require participants to be physically located in the U.S. and to speak English as a first language. For each human participant study, we source a distinct set of participants (no overlap) to avoid potential bias in annotations that could occur from participating in related studies in the past. For each study, we measure the average inter-annotator agreement using Fleiss’s kappa (Fleiss, 1971).

We choose ChatGPT (OpenAI, 2022) (gpt-3.5-turbo-0613) as the LLM we use in the implementation. As mentioned in section 3, our method does not rely on any external story corpora. Suspenseful stories can be generated with our method in a fully zero-shot way from ChatGPT. We start every ChatGPT dialogue session with “You are a creative storyteller.”. When generating anything other than stories (e.g., interim information such as the protagonist’s goal and actions), to discourage ChatGPT from

generating excessively long responses that are usually not quite useful when serving as inputs to subsequent generation stages, we append “Use no more than $[n]$ sentences in your answer.” (n can be three, four, or five) to the prompt when necessary.

4.1 Comparison Against Baseline Method

In this study, we seek to understand whether our method has achieved improvement in terms of story suspense as well as other typical story evaluation criteria. We choose the following method as the baseline to compare with: directly asking ChatGPT for a suspenseful story with straightforward prompts. To make it a fair comparison, we control that the stories are generated conditioned on the same background information (the protagonist’s name, occupation, and goal) used in our method’s generation process. The prompt we use for generating the baseline stories is: “Story Background: \$name_occupation. \$goal. Write a full suspenseful story based on the story background.” (variables \$name_occupation and \$goal are obtained from subsection 3.1).

We recruited 90 participants to evaluate 30 pairs of stories, with each individual evaluating 10 pairs of stories, and each pair of stories was evaluated by 30 participants. Each pair of stories comprises one story generated by our method and one story generated by the baseline method. Both stories are randomly selected from a corresponding large collection of stories generated with either our method or the baseline method.

We ask human participants to make a pair-wise comparison and assess the goodness of pairs of suspenseful stories from the following dimensions:

- **Suspense:** “Which story makes you feel a higher level of suspense?”;
- **Novelty:** “Which story do you find to be more novel?”;
- **Enjoyment:** “Which story is more enjoyable to read?”;
- **Logical Sense:** “Which story logically makes more sense?”;
- **Naturalness:** “Which story is more likely to have been written by a human?”.

Results are presented in Table 3 where we show the percentage of times stories generated from each method are preferred for each evaluation metric. Participants’ preferences with a fair average inter-annotator agreement show that compared to the baseline, our method significantly improves the

²<https://www.prolific.co/>

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	84.9**	11.8	3.3	76.6**	13.6	9.8	67.9**	19.4	12.7	49.2*	27.8	23.0	25.3	18.6	56.1**
Ours v.s. Ablation #1	75.0**	19.2	5.8	63.1**	16.5	20.4	59.9**	17.2	22.9	37.8	27.1	35.1	31.0	29.9	39.1
Ablation #1 v.s. Baseline	79.8**	12.9	7.3	68.9**	11.3	19.8	71.7**	12.1	16.2	45.3	30.9	23.8	33.2	28.6	38.2
Ours v.s. Ablation #2	80.4**	9.9	9.7	66.0**	10.3	23.7	68.6**	16.6	14.8	39.4	32.4	28.2	22.8	24.0	53.2**
Ablation #2 v.s. Baseline	72.3**	12.7	15.0	74.3**	9.9	15.8	63.1**	15.3	21.6	29.6	22.3	48.1*	30.9	25.4	43.7

Table 3: Human evaluation results for both the comparison against the ChatGPT baseline and the ablation studies, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ (* at $p < 0.05$) confidence level using a Wilcoxon sign test.

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	73.9**	15.7	10.4	66.2**	9.6	24.2	59.8**	18.3	21.9	42.6	23.6	33.8	25.1	19.9	55.0**

Table 4: Human evaluation results for the comparison against the Llama 2 baseline, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ confidence level using a Wilcoxon sign test.

human perception of narrative suspense, and effectively produces more novel and enjoyable stories. Our method also achieves improvement in the logical sense of stories. We attribute this to our method’s outline planning component which can make the narratives more rational and reasonable. As our method and the baseline method both use ChatGPT, the naturalness of our stories stays at the same level as the baseline’s.

To further validate the effectiveness of our method across different LLMs, we perform an additional experiment on Llama 2 (Touvron et al., 2023) (Llama-2-13b-chat). We follow the same experimental setup as introduced before, except changing the LLM from ChatGPT to Llama 2. Results presented in Table 4 show that the Llama 2 augmented with our method can generate more suspenseful, more novel, and more enjoyable stories compared to a vanilla Llama 2. These results echo the main results reported in Table 3 and therefore prove that our method is stable and transferrable across different LLMs.

Since the stories generated by our method are long-form (see Appendix C for an example), we include Re3 (Yang et al., 2022b), a commonly used long-form story generation technique, as an additional baseline to compare with. We condition both Re3 and our method on the exact same background information to generate full stories. Results are presented in Table 5, showing that our method can generate more suspenseful, more novel, and more enjoyable stories compared to Re3.

4.2 Ablation Studies

To fully investigate the effectiveness of our method and confirm the necessity of two components: outline planning (§3.2) and detail elaboration (§3.3), we perform two corresponding ablation studies. First, to study the necessity of detail elaboration, we ablate our method (denoted as Ablation #1) by replacing the detail elaboration component with a straightforward prompt as presented in Table 6. Second, to study the necessity of outline planning, based on Ablation #1 we ablate our method (denoted as Ablation #2) by further replacing the outline planning component with a similarly straightforward prompt (also listed in Table 6). We recruited 90 participants and followed the evaluation setup of subsection 4.1. Participants’ preferences presented in Table 3 with a fair average inter-annotator agreement show that both outline planning (§3.2) and detail elaboration (§3.3) are necessary for generating novel and enjoyable suspenseful stories.

4.3 Examination of Story Outline

After confirming the effectiveness of our method and the necessity of individual method components, we need to further examine the story outline generated in subsection 3.2 to determine if our method’s internal workflow is running properly as we expected. We asked (1) whether the actions of the protagonist are relevant to the goal, (2) whether a plausible reason is given for the protagonist’s failure, and (3) whether each of the protagonist’s actions, after each failure, is perceived as decreas-

Methods	Suspense			Novelty			Enjoyment			Logical Sense			Naturalness		
	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%	Win%	Lose%	Tie%
Ours v.s. Baseline	81.1**	6.8	12.1	84.7**	11.3	4.0	70.8**	11.9	17.3	33.1	25.7	41.2	28.4	12.0	59.6**

Table 5: Human evaluation results for the comparison against the Re3 baseline, showing the percentage of times stories generated from each method are preferred for five evaluation metrics. “Win” means the method on the left is preferred. ** indicates results are significant at $p < 0.01$ confidence level using a Wilcoxon sign test.

Ablation #1’s Prompt
Story Background: \$name_occupation. \$goal. Action #1: \$action1 Reason #1: \$reason1 Action #2: \$action2 Reason #2: \$reason2 Action #3: \$action3 Story Summary: Aiming to achieve the goal: \$goal, the protagonist first tries to take Action #1, but fails due to Reason #1; then tries to take Action #2, but fails due to Reason #2; finally tries to take Action #3 and this time the protagonist successfully achieves the goal. Write a full suspenseful story based on the story summary.
Ablation #2’s Prompt
Story Background: \$name_occupation. \$goal. Story Summary: Aiming to achieve the goal: \$goal, the protagonist first tries to take a first action, but fails due to a reason; then tries to take another new action, but fails due to another reason; finally tries to take a new action and this time the protagonist successfully achieves the goal. Write a full suspenseful story based on the story summary.

Table 6: The prompts we use in ablation studies. All the variables shaped as \$variable are obtained in [subsection 3.1](#) and [subsection 3.2](#), as reflected in [Table 1](#) and [Table 2](#).

ing in likelihood.

For each aspect, we recruited 30 participants to examine the goal, actions, and reasons for failure of 30 randomly selected stories generated by our method. On the question of **protagonist action related to goal**, participants reported that protagonist actions were relevant to the goal 96.5% of the time. Regarding **plausible reasons for failure**, 89.8% of the time. We find that in the vast majority of cases, the action is relevant to the goal and the reason is plausible for the protagonist’s failure. On the question of **decreasing likelihood**, 55.1% of participants indicated that they thought the likelihood of actions was going down, 13.9% indicated that they thought the likelihood was going up, and 31.0% thought the likelihood stayed the same. More than half of participants perceived declining likelihood trends, and a clear plurality felt that likelihoods were decreasing or staying the

same. These results all indicate that the internal workflow of our method is running properly as we expected.

5 Factors Affecting Suspense Perception

Given the success of our method in creating suspenseful stories, we recognized a unique opportunity to contribute to a more fine-grained understanding of suspense. Prior work from psychology used hand-crafted stories from literature and thus did not have the means to run controlled studies with a large number of stories with fine-grained interventions. We conducted a study to investigate the timing and ordering of knowledge revealed to the reader, as well as the relationship between empathy for the protagonist and suspense. We specifically investigate how the following affect suspense perception: (1) clues that set up the failure; (2) information revelation timing (before vs after); (3) reader’s empathy.

Clue setup: we recruited 90 participants to ask for their preferences regarding the suspense level of 30 randomly selected pairs of stories. For each pair of stories, which is generated with and without clue setup (detailed in [subsection 3.3](#)) respectively, we ask participants which one of the two stories makes them feel a higher level of suspense. 57.9% of participants reported higher suspense when reading stories with clues, whereas 10.9% of participants reported higher suspense when reading stories without clues (31.2% were a tie). These results are significant at $p < 0.05$ confidence level using a Wilcoxon sign test. Thus we conclude that clues of impending failure are a factor that increases reader perceptions of suspense.

Information revelation timing: we recruited 90 participants to ask for their preferences regarding the suspense level of 30 randomly selected pairs of stories. For each pair of stories, which is generated by revealing the information *beforehand* and *after the fact* (detailed in [subsection 3.3](#)) respectively, we ask participants which one of the two stories makes them feel a higher level of suspense. 42.1% of participants reported higher suspense when read-